

Onshape Workflow Manual

Wind Turbine Design Lab exports an Onshape package so students can rebuild a selected blade in CAD before creating an STL for 3D printing. This manual walks through the workflow slowly, assuming the user is new to Onshape.

This export is not a finished STL. It is a CAD reference package: it gives the blade table, 2D outlines, and airfoil section profiles. Students still need to build the 3D part in Onshape, check the hub connection, and export STL from Onshape.

1. Files in the downloaded ZIP

Click **Download Onshape package** in the app, then unzip the file. You should see these files:

File	Use
blade_geometry.csv	Main table: section, position, chord, twist, airfoil, and role.
airfoil_sections.csv	Airfoil metadata and recommended Reynolds-number ranges.
blade_planform.dxf	Top-view blade outline in centimetres. Use it as an outline check.
section_profiles.dxf	Airfoil cross-section sketches for each blade station.
design_metadata.json	Simulator settings saved with the export.
onshape_build_guide.md	Short build notes included inside the export.

For the first Onshape build, keep `blade_geometry.csv` open beside Onshape. It is the map for where every airfoil section belongs.

2. Read the geometry table first

Open `blade_geometry.csv`. The important columns are:

Column	Meaning
Section	Root-to-tip section number.
Position (cm)	Distance from the turbine center along the blade radius.
Chord (cm)	Airfoil width from leading edge to trailing edge.
Twist (deg)	Rotation angle for that section.
Airfoil	Airfoil profile assigned to that section.
Role	Plain-language reason for using that airfoil there.

Example:

```
Section,Position (cm),Chord (cm),Twist (deg),Airfoil
1 (Root),6.0,9.0,20.0,NACA 4418
2,14.0,7.5,15.0,SG6040
3,24.0,5.8,10.0,SG6043
4,34.0,4.2,6.0,SG6043
5,43.0,3.0,2.0,NACA 2412
6 (Tip),50.0,1.8,0.0,E387
```

The table positions are measured from the rotor center. In CAD it is usually easier to start the blade at the first section. If section 1 is at 6 cm and section 6 is at 50 cm, then the CAD station offsets from section 1 are:

Section	Table position	CAD offset from section 1
1	6 cm	0 cm
2	14 cm	8 cm
3	24 cm	18 cm
4	34 cm	28 cm
5	43 cm	37 cm
6	50 cm	44 cm

Formula:

$\text{CAD offset} = \text{section position} - \text{root section position}$

3. Create a new Onshape document

1. Open Onshape.
2. Create a new Document.
3. Open a Part Studio.
4. Use centimetres as the working unit when importing DXF files.

If a sketch imports much too large or much too small, the unit was probably wrong. Delete the import and insert it again with centimetres.

4. Import the top-view planform

The planform is a top-view blade outline. It is useful as a visual guide, but it is not the main Loft input.

1. Start a Sketch on the Top plane.
2. Insert or import `blade_planform.dxf`.
3. Choose centimetres if Onshape asks for units.
4. Keep this sketch visible as a reference.

Use this file to check whether the final blade outline looks similar to the simulator's top-view design.

5. Create section planes

Create one plane for each blade section. The planes should be perpendicular to the blade length direction.

For the example table above:

Plane name	Offset from first/root section
S1 Root	0 cm
S2	8 cm
S3	18 cm
S4	28 cm
S5	37 cm
S6 Tip	44 cm

The actual numbers must come from your own `blade_geometry.csv`.

Tip: rename each plane immediately. Clear names make Loft much easier.

6. Place airfoil profiles on the planes

`section_profiles.dxf` contains all exported airfoil profiles. For a beginner, the safest workflow is:

1. Create a Sketch on plane S1 Root.
2. Insert `section_profiles.dxf`.
3. Keep only the section 1 profile.
4. Delete the other imported profiles from that sketch.
5. Move the remaining profile so its centreline or leading edge is on your blade

reference axis.

6. Rotate it by the section 1 Twist angle from `blade_geometry.csv`.
7. Finish the sketch.
8. Repeat for section 2, section 3, and so on.

If the DXF import is difficult to separate, use the text labels inside the DXF. Each profile is labelled with its section, airfoil, chord, and twist.

7. Apply twist correctly

Twist is the local rotation of each airfoil section.

For each sketch:

Section	Rotate profile by
Section 1	Twist from row 1
Section 2	Twist from row 2
Section 3	Twist from row 3
Section 4	Twist from row 4
Section 5	Twist from row 5
Section 6	Twist from row 6

Use the values in your CSV, not the example values if your design is different.

Keep the rotation direction consistent. If the Loft appears twisted backward, undo, reverse the rotation direction, and try again.

8. Align the profiles before Loft

Loft works best when all profiles share a consistent reference point.

Pick one alignment method and use it for every section:

- align each profile by chord midpoint; or
- align each profile by leading edge; or
- align each profile by trailing edge.

For classroom blades, chord midpoint alignment is usually easiest. Leading-edge alignment can also work, but do not mix alignment methods section by section.

Before Loft, check:

- every sketch has exactly one closed airfoil profile;
- profiles are ordered from root to tip;
- the same side of the airfoil faces the same direction on every plane;
- twist is applied in the intended direction.

9. Create the 3D blade with Loft

1. Start the Loft feature.
2. Choose **Solid** if each profile is a closed region.
3. Select the section sketches in order:
 - section 1 root profile; - section 2 profile; - section 3 profile; - section 4 profile; - section 5 profile; - section 6 tip profile.
4. Preview the shape.
5. If the blade twists strangely, cancel and check profile alignment and order.
6. Accept the Loft.

If Solid Loft fails, try Surface Loft first. If Surface Loft works but Solid Loft does not, at least one profile may not be a clean closed region.

10. Add the hub connection

The simulator exports the blade geometry, not the final hub. Students must add a real mount that matches their generator shaft and competition rules.

Common classroom approach:

1. Create a cylinder for the hub.
2. Add holes or shaft adapter geometry to match the generator.
3. Extend or thicken the blade root so it joins the hub safely.
4. Fillet the root connection if possible.
5. Check that the full rotor diameter does not exceed the rule limit.

For a 1 m diameter rule, the final rotor radius must be no more than 50 cm from the center to the farthest blade tip.

11. Create three blades

After one blade and hub connection are correct:

1. Use Circular Pattern.
2. Select the blade body or blade feature.
3. Use the hub axis as the rotation axis.
4. Set instance count to 3.
5. The blades should be spaced 120 degrees apart.

Check that the three-blade pattern does not overlap the hub, shaft, or frame.

12. Export STL from Onshape

When the CAD model is ready:

1. Right-click the finished Part or Assembly.
2. Choose Export.

3. Select STL.
4. Use millimetres if your slicer expects millimetres.
5. Open the STL in your slicer.
6. Check print orientation, supports, wall thickness, and root strength.

Always inspect the STL before printing. A successful Onshape export does not guarantee the part is strong enough for testing.

13. Troubleshooting

Problem	Likely cause	Fix
DXF imports at the wrong size	Wrong import unit	Reimport using centimetres.
Loft fails	Profile is not closed or contains extra curves	Clean the sketch; leave one closed profile per section.
Loft twists into a knot	Profiles selected out of order or aligned inconsistently	Select root to tip; align all profiles by the same reference point.
Blade looks mirrored	Profile orientation is flipped on one or more planes	Flip or rotate the affected sketch.
Blade outline does not match planform	Profiles are shifted sideways	Use <code>blade_planform.dxf</code> as a top-view reference and realign profiles.
Final part is too large	Rotor diameter or unit conversion issue	Check radius and units before exporting STL.
Root breaks easily	Hub connection too thin	Add root reinforcement, fillets, or a thicker hub mount.

14. Recommended beginner workflow

For the first successful build, do not try to optimize everything.

1. Load the **Easy CAD / Easy Print** preset in the simulator.
2. Download the Onshape package.
3. Build only one blade first.
4. Use chord midpoint alignment for every section.
5. Loft the blade.
6. Add a simple hub connector.
7. Export STL.
8. Print one small test if possible.
9. Return to the simulator and improve one variable at a time.

15. What the simulator does not do yet

The current project does not automatically create:

- a finished Onshape document;
- a complete parametric CAD model;
- a print-ready STL;

- supports, slicer settings, or infill settings;
- structural safety verification.

The export is intended to reduce manual copying errors and make CAD rebuilding more consistent. Real wind-tunnel testing is still required for final calibration.

References

- Onshape Help: Insert DXF or DWG into a sketch -
<https://cad.onshape.com/help/Content/Sketch/insert_dwg_or_dxf.htm>
- Onshape Help: Loft - <<https://cad.onshape.com/help/Content/PartStudio/loft.htm>>